

Lecture 6: Basics of Data Communication

◆ Key Terminologies

- **Data:** Raw facts and figures such as text, numbers, images, audio, and video.
Example: A list of student names and marks before processing is data.
- **Data Communication:** Transfer of data electronically between devices.
Example: Sending an email from a laptop to a smartphone.
- **Data Transmission:** Sending data via wired or wireless medium.
Example: Streaming a YouTube video over Wi-Fi.
- **Analog Signals:** Continuous waves that vary with time, prone to interference.
Example: Human voice in a phone call, represented by sine waves.
- **Digital Signals:** Discrete binary values (0s and 1s), less prone to noise.
Example: Data stored on a USB drive, represented by square waves.
- **Data Rate (Bit Rate):** Number of bits transmitted per second.
Example: A 100 Mbps internet connection transfers 100 million bits per second.
- **Baud Rate:** Number of signal units per second; one signal may represent multiple bits. *Example:* A modem transmitting at 1200 baud means 1200 signal changes per second.
- **Signal-to-Noise Ratio (SNR):** Ratio of signal power to noise power, measured in decibels.
Example: A high SNR in a Wi-Fi connection means clearer data transmission.

◆ Components of a Communication System

1. **Message** – The information to be sent (text, audio, video).
Example: A WhatsApp message containing text and emojis.
2. **Sender** – Device generating the message.
Example: A computer sending an email.

3. **Receiver** – Device receiving the message.
Example: A smartphone receiving the email.
4. **Medium** – Path for transmission (wired or wireless).
Example: Fiber optic cable for internet, or radio waves for mobile phones.
5. **Protocol** – Rules for communication.
Example: TCP/IP protocol used in the internet.

◆ Properties of a Good Communication System

- **Delivery** – Data must reach the correct destination. *Example:* An online banking transaction must reach the correct account.
- **Accuracy** – Data must be error-free. *Example:* A corrupted file download is useless.
- **Timeliness** – Data must arrive on time. *Example:* Live video streaming must deliver frames without delay.

◆ Transmission Media

- **Guided (Wired):**
 - *Twisted Pair:*
 - **UTP** – Cheap, used in telephone lines.
 - **STP** – Shielded, used in Ethernet networks.
 - *Coaxial Cable:* Used in cable TV connections.
 - *Fiber Optic:* Uses light, very high speed, used in modern internet backbones.
- **Unguided (Wireless):**
 - *Radio Waves:* Omnidirectional, used in FM/AM radios, cordless phones.
 - *Microwaves:* Line-of-sight, used in mobile towers and satellite TV.
 - *Infrared:* Short-range, used in TV remote controls and wireless LANs.

◆ Transmission Impairments

- **Attenuation:** Loss of signal strength over distance.

Example: Wi-Fi signal weakens as you move farther from the router.

- **Distortion:** Signal shape changes due to varying speeds of components.

Example: Audio distortion in long-distance phone calls.

- **Noise:** Unwanted signals interfering with transmission.

Example: Crosstalk in telephone lines or static in radio signals.

30 multiple-choice questions (MCQs)

MCQs: Basics of Data Communication

◆ Section 1: Key Terminologies

1. **Which of the following is considered raw facts and figures?**

- a) Information
- b) Data
- c) Protocol
- d) Message

2. **Sending an email from a laptop to a smartphone is an example of:**

- a) Data Transmission
- b) Data Communication
- c) Analog Signal
- d) Noise

3. Streaming a YouTube video over Wi-Fi is an example of:

- a) Data Transmission
- b) Protocol
- c) Distortion
- d) Baud Rate

4. Human voice in a phone call is an example of:

- a) Digital Signal
- b) Analog Signal
- c) Noise
- d) Protocol

5. Data stored on a USB drive is represented by:

- a) Analog Signals
- b) Digital Signals
- c) Noise
- d) Distortion

6. **A 100 Mbps internet connection transfers:**
- a) 100 million bytes per second
 - b) 100 million bits per second
 - c) 100 thousand bits per second
 - d) 100 signals per second
7. **A modem transmitting at 1200 baud means:**
- a) 1200 bits per second
 - b) 1200 signal changes per second
 - c) 1200 bytes per second
 - d) 1200 protocols per second
8. **Signal-to-noise ratio (SNR) is measured in:**
- a) Hertz
 - b) Bits
 - c) Decibels
 - d) Bytes
-

◆ Section 2: Components of Communication System

9. Which of the following is the information to be sent?
- a) Sender
 - b) Receiver
 - c) Message
 - d) Protocol
10. A computer sending an email acts as:
- a) Receiver
 - b) Sender
 - c) Medium
 - d) Protocol
11. A smartphone receiving an email acts as:
- a) Sender
 - b) Receiver
 - c) Medium
 - d) Protocol
12. Fiber optic cable used for internet is an example of:
- a) Message
 - b) Medium

- c) Protocol
 - d) Noise
13. TCP/IP is an example of:
- a) Message
 - b) Medium
 - c) Protocol
 - d) Noise

◆ Section 3: Properties of Good Communication

14. An online banking transaction reaching the correct account shows:
- a) Accuracy
 - b) Delivery
 - c) Timeliness
 - d) Noise
15. A corrupted file download violates which property?
- a) Delivery
 - b) Accuracy
 - c) Timeliness
 - d) Protocol
16. Live video streaming requires:
- a) Delivery
 - b) Accuracy
 - c) Timeliness
 - d) Noise

◆ Section 4: Transmission Media

17. Telephone lines commonly use:
- a) Coaxial Cable
 - b) Fiber Optic Cable
 - c) Twisted Pair Cable
 - d) Infrared
18. Cable TV connections use:
- a) Twisted Pair Cable
 - b) Coaxial Cable

- c) Fiber Optic Cable
 - d) Radio Waves
19. Modern internet backbones use:
- a) Twisted Pair Cable
 - b) Coaxial Cable
 - c) Fiber Optic Cable
 - d) Infrared
20. FM/AM radios use:
- a) Radio Waves
 - b) Microwaves
 - c) Infrared
 - d) Fiber Optic
21. Mobile towers and satellite TV use:
- a) Radio Waves
 - b) Microwaves
 - c) Infrared
 - d) Coaxial Cable
22. TV remote controls use:
- a) Radio Waves
 - b) Microwaves
 - c) Infrared
 - d) Fiber Optic

◆ Section 5: Transmission Impairments

23. Wi-Fi signal weakening with distance is an example of:
- a) Distortion
 - b) Noise
 - c) Attenuation
 - d) Protocol
24. Audio distortion in long-distance calls is an example of:
- a) Noise
 - b) Distortion
 - c) Attenuation
 - d) Delivery

25. Crosstalk in telephone lines is an example of:
- a) Noise
 - b) Distortion
 - c) Attenuation
 - d) Protocol
26. Which impairment refers to change in signal shape?
- a) Noise
 - b) Distortion
 - c) Attenuation
 - d) Delivery
27. Which impairment refers to unwanted signals mixing with original signals?
- a) Noise
 - b) Distortion
 - c) Attenuation
 - d) Delivery
28. Which impairment is measured in decibels?
- a) Noise
 - b) Distortion
 - c) Attenuation
 - d) Protocol
29. Amplifiers are used to overcome:
- a) Noise
 - b) Distortion
 - c) Attenuation
 - d) Protocol
30. Different signal components arriving at different times cause:
- a) Noise
 - b) Distortion
 - c) Attenuation
 - d) Delivery
-

✓ Answers with Explanations

1. **b) Data** – Raw facts and figures are called data.
2. **b) Data Communication** – Email transfer is communication between devices.
3. **a) Data Transmission** – Streaming involves sending data over a medium.
4. **b) Analog Signal** – Human voice is continuous.
5. **b) Digital Signals** – USB stores data in binary.
6. **b) 100 million bits per second** – Mbps = megabits per second.
7. **b) 1200 signal changes per second** – Baud rate counts signal changes.
8. **c) Decibels** – SNR is measured in dB.
9. **c) Message** – Information to be sent.
10. **b) Sender** – Computer sending email.
11. **b) Receiver** – Smartphone receiving email.
12. **b) Medium** – Fiber optic is transmission medium.
13. **c) Protocol** – TCP/IP is a communication rule set.
14. **b) Delivery** – Correct destination shows delivery.
15. **b) Accuracy** – Corrupted data violates accuracy.
16. **c) Timeliness** – Streaming requires timely delivery.
17. **c) Twisted Pair Cable** – Telephone lines use twisted pair.
18. **b) Coaxial Cable** – Cable TV uses coaxial.
19. **c) Fiber Optic Cable** – Internet backbone uses fiber optic.
20. **a) Radio Waves** – Radios use radio waves.
21. **b) Microwaves** – Mobile towers use microwaves.
22. **c) Infrared** – TV remotes use infrared.
23. **c) Attenuation** – Signal weakens with distance.
24. **b) Distortion** – Signal shape changes cause distortion.
25. **a) Noise** – Crosstalk is unwanted signal interference.
26. **b) Distortion** – Change in signal shape.
27. **a) Noise** – Unwanted signals mixing.
28. **c) Attenuation** – Measured in decibels.
29. **c) Attenuation** – Amplifiers boost weakened signals.
30. **b) Distortion** – Different arrival times cause distortion.

Lecture 7: Transmission Medium

Transmission Medium

Definition

- Transmission Medium (Communication Channel): Path (wireless or physical) between sender and receiver for data transfer.
 - Data transmitted via **electromagnetic/electrical signals** through wires, atmosphere, or vacuum.
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✦ Characteristics of Transmission

1. **Delivery** – Data must reach the correct destination in the correct order.
 2. **Accuracy** – Data must be delivered without alteration.
 3. **Timeliness** – Data must arrive on time; late data is useless.
-

✕ Types of Transmission Media

Broadly classified into:

- **Guided Media (Wired/Bounded)**
 - **Unguided Media (Wireless/Unbounded)**
-

4.2.1 Guided Media

Signals transmitted via physical links (cables).

Types:

1. Twisted Pair Cable

- Two insulated copper wires twisted together.
- Prevents external interference.
- Widely used in data/voice infrastructures.
- Types:
 - **UTP (Unshielded Twisted Pair):**
 - Blocks some interference but vulnerable.
 - Used in telephones/less distant network PCs.
 - Cheap, easy to install.
 - **STP (Shielded Twisted Pair):**
 - Special coating to block interference.

- Used in fast Ethernet, telephone voice/data channels.

2. Coaxial Cable (Coax)

- Outer plastic covering with two parallel conductors.
- Used in cable TV, analog television networks.

3. Fiber-Optic Cable

- Data transmitted as light signals.
- Core (glass/plastic) with cladding for reflection.
- Very high speed, large data volumes.
- Used in modern high-speed networks.

4.2.2 Unguided Media

Wireless transmission; no physical medium.

Types:

1. Radio Waves

- Electromagnetic waves, omnidirectional.
- Easy to generate, penetrate buildings.
- Used in FM/AM radio, TV, cordless phones.

2. Microwaves

- Line-of-sight transmission; antennas must align.
- Distance $\propto$ antenna height.
- Used in mobile communication towers, TV broadcast.
- Types:
 - **Terrestrial:** Antennas on Earth.
 - **Satellite:** One antenna in orbit, others on Earth.

3. Infrared

- Uses infrared light (LED transmitters, photodiode receivers).
- Line-of-sight, short range.
- Cannot penetrate walls.
- Used in wireless LANs, short-range links.

4.2.3 Transmission Impairments

Defects during data transmission.

Causes:

1. Attenuation

- Loss of signal energy due to medium resistance.
- Increases with distance.
- Measured in decibels.
- Amplifiers used to restore strength.

2. **Distortion**

- Change in signal shape.
- Composite signals' components arrive at different times due to varying speeds.
- Causes phase differences.

3. **Noise**

- Unwanted signals mixing with original data.
- Types: induced, crosstalk, thermal, impulse.
- Can damage signals.

Key Learning Outcomes (SLOs)

- Understand transmission medium.
- Recognize and describe guided vs. unguided media.
- Differentiate between them.
- Identify flaws/faults in transmission signals (attenuation, distortion, noise).

MCQs on Transmission Medium

1.

Transmission medium refers to:

- A) Only wireless communication
 - B) Path between sender and receiver for data transfer
 - C) Storage devices
 - D) Internet protocols
-

2.

Which is **NOT** a fundamental characteristic of transmission?

- A) Delivery
 - B) Accuracy
 - C) Timeliness
 - D) Encryption
-

3.

Transmission media are broadly classified into:

- A) Digital and Analog
 - B) Guided and Unguided
 - C) Wired and Wireless only
 - D) Optical and Electrical
-

4.

Guided media is also called:

- A) Wireless media
 - B) Bounded or wired media
 - C) Satellite media
 - D) Optical media
-

5.

Which of the following is **NOT** a guided medium?

- A) Twisted pair cable
 - B) Coaxial cable
 - C) Fiber-optic cable
 - D) Radio waves
-

6.

Twisted pair cables are made of:

- A) Aluminum wires

- B) Insulated copper wires twisted together
 - C) Fiber glass
 - D) Plastic conductors
-

7.

Which type of twisted pair cable is cheaper and easier to install?

- A) Shielded Twisted Pair (STP)
 - B) Unshielded Twisted Pair (UTP)
 - C) Fiber-optic cable
 - D) Coaxial cable
-

8.

STP cables are mainly used in:

- A) Telephonic applications
 - B) Fast-data-rate Ethernet and telephone lines
 - C) Cable TV networks
 - D) Satellite communication
-

9.

Coaxial cables are widely used in:

- A) Cable TV and analog television networks
 - B) Optical fiber communication
 - C) Mobile towers
 - D) Satellite systems
-

10.

Fiber-optic cables transmit data in the form of:

- A) Electrical signals
 - B) Radio waves
 - C) Light signals
 - D) Sound waves
-

11.

Unguided media is also called:

- A) Wired media
 - B) Wireless or unbounded media
 - C) Optical media
 - D) Digital media
-

12.

Which of the following is **NOT** an unguided medium?

- A) Radio waves
 - B) Microwaves
 - C) Infrared
 - D) Coaxial cable
-

13.

Radio waves are:

- A) Directional
 - B) Omnidirectional
 - C) Line-of-sight only
 - D) Unable to penetrate buildings
-

14.

Microwave transmission requires:

- A) Omnidirectional antennas
 - B) Line-of-sight alignment between antennas
 - C) Fiber-optic cables
 - D) Coaxial cables
-

15.

Which type of microwave transmission uses antennas on Earth only?

- A) Satellite microwave
 - B) Terrestrial microwave
 - C) Infrared microwave
 - D) Fiber microwave
-

16.

Infrared transmission cannot:

- A) Use LEDs and photodiodes
 - B) Penetrate walls or objects
 - C) Be used for short-range communication
 - D) Operate at terahertz frequency
-

17.

Which of the following is a **short-range wireless link** technology?

- A) Fiber-optic cable
- B) Infrared communication

- C) Coaxial cable
 - D) Satellite microwave
-

18.

Transmission impairments refer to:

- A) Encryption of signals
 - B) Loss of signal quality during transmission
 - C) Increase in bandwidth
 - D) Use of amplifiers
-

19.

Attenuation means:

- A) Signal distortion
 - B) Loss of energy during transmission
 - C) Addition of noise
 - D) Increase in frequency
-

20.

Distortion occurs because:

- A) Signal components travel at different speeds and arrive out of phase
 - B) Noise mixes with the signal
 - C) Amplifiers are not used
 - D) Fiber-optic cables are broken
-
-

✓ Answers with Explanations

1. **B** – Transmission medium is the path between sender and receiver.
2. **D** – Encryption is not listed as a fundamental characteristic.
3. **B** – Media are classified into guided and unguided.
4. **B** – Guided media is bounded/wired.
5. **D** – Radio waves are unguided.
6. **B** – Twisted pair uses insulated copper wires.
7. **B** – UTP is cheaper and easier to install.
8. **B** – STP is used in fast Ethernet and telephone lines.
9. **A** – Coaxial cables are used in cable TV and analog TV.
10. **C** – Fiber-optic cables transmit data as light signals.
11. **B** – Unguided media = wireless/unbounded.
12. **D** – Coaxial cable is guided, not unguided.
13. **B** – Radio waves are omnidirectional.

- 14.**B** – Microwaves require line-of-sight alignment.
- 15.**B** – Terrestrial microwaves use Earth-based antennas.
- 16.**B** – Infrared cannot penetrate walls.
- 17.**B** – Infrared is used for short-range wireless links.
- 18.**B** – Impairments = loss of signal quality.
- 19.**B** – Attenuation = loss of energy.
- 20.**A** – Distortion occurs when signal components arrive at different times/phases.

Lecture 8: Communication Devices



Communication Devices

Definition

- A **communication device** is hardware capable of **transmitting and receiving data, instructions, and information**.
-

🔌 Types of Communication Devices

4.3.1 Switch

- **Network switch** connects computers and other devices (printers, scanners, cameras) on a network.
 - Data cables from all devices plug into the switch.
 - Enables communication between connected devices.
-

4.3.2 Router

- Connects **two or more networks**.
 - Combination of hardware and software.
 - Main function: **determine optimal data path** and transfer information (acts as a **network traffic controller**).
-

4.3.3 Modem

- **Modem = Modulator + Demodulator**.
- **Modulation**: Converts digital signals → analog.
- **Demodulation**: Converts analog signals → digital.
- Enables computers to share information via phone lines, cables, or satellites.
- **Types of Modems**:
 1. **Dial-up Modem**
 - Uses standard telephone lines.
 - Converts digital packets into analog signals.
 - Can be internal or external.
 - Slow compared to modern technologies.
 2. **DSL Modem (Digital Subscriber Line)**
 - Uses telephone lines like dial-up.
 - Built-in network switch.
 - Transfers data/voice at **higher speed**.
 - Some support wireless communication.
 3. **ISDN Modem (Integrated Services Digital Network)**
 - Digital phone connection.
 - Transmits **data, voice, video simultaneously** over telephone lines.

- Faster but more expensive.
 - Converts analog voice → digital signals before transmission.
-

4.3.4 Network Interface Card (NIC)

- Hardware device connecting a computer to a network.
 - Installed on the **motherboard**.
 - Establishes physical connection between computer and network.
 - Translates computer data into electrical signals for transmission.
 - Modern motherboards often have **built-in NICs**.
-

Key Learning Outcomes (SLOs)

- Define communication devices.
- Explain router functions.
- Differentiate types of modems (Dial-up, DSL, ISDN).
- Describe functions of switch, router, NIC.
- Elaborate functions of communication devices in networking.

MCQs on Communication Devices

1.

A communication device is defined as:

- A) Software for data storage
 - B) Hardware capable of transmitting and receiving data
 - C) A type of operating system
 - D) A wireless protocol
-

2.

Which device connects computers and peripherals like printers and scanners within a network?

- A) Router
 - B) Switch
 - C) Modem
 - D) NIC
-

3.

The main function of a router is:

- A) Converting signals
 - B) Determining optimal data path and controlling traffic
 - C) Connecting devices within a LAN
 - D) Translating data into electrical signals
-

4.

A router is a combination of:

- A) Hardware only
 - B) Software only
 - C) Hardware and software
 - D) None of the above
-

5.

Modem stands for:

- A) Modern Emulator
 - B) Modulator and Demodulator
 - C) Modular Device
 - D) Multi-data Emulator
-

6.

Modulation refers to:

- A) Converting analog signals into digital
 - B) Converting digital signals into analog
 - C) Encrypting signals
 - D) Compressing signals
-

7.

Demodulation refers to:

- A) Converting digital signals into analog
 - B) Converting analog signals into digital
 - C) Splitting signals
 - D) Amplifying signals
-

8.

Which medium can modems use to share information?

- A) Phone lines
 - B) Cables
 - C) Satellite connections
 - D) All of the above
-

9.

Dial-up modems use:

- A) Fiber-optic cables
 - B) Standard telephone lines
 - C) Coaxial cables
 - D) Wireless signals only
-

10.

Telephone lines carry:

- A) Digital signals
 - B) Analog signals
 - C) Both digital and analog simultaneously
 - D) Optical signals
-

11.

DSL stands for:

- A) Digital Subscriber Line
- B) Data Signal Line

- C) Direct Service Link
 - D) Digital Switching Line
-

12.

DSL modems differ from dial-up modems because they:

- A) Use fiber optics
 - B) Have built-in network switches and higher speed
 - C) Cannot use telephone lines
 - D) Only transmit voice
-

13.

Some DSL modems also support:

- A) Satellite communication
 - B) Wireless communication
 - C) Fiber-optic transmission
 - D) Infrared signals
-

14.

ISDN stands for:

- A) Integrated Services Digital Network
 - B) Internal Signal Data Network
 - C) International Subscriber Digital Network
 - D) Integrated Signal Distribution Node
-

15.

ISDN modems can transmit:

- A) Only data
 - B) Only voice
 - C) Data, voice, and video simultaneously
 - D) Only video
-

16.

ISDN technology is described as:

- A) Slow and inexpensive
 - B) Faster but more expensive
 - C) Wireless only
 - D) Limited to analog signals
-

17.

Network Interface Card (NIC) is installed on:

- A) Hard drive
 - B) Motherboard
 - C) Router
 - D) Switch
-

18.

The main function of NIC is:

- A) Encrypting data
 - B) Establishing physical connection between computer and network
 - C) Acting as a traffic controller
 - D) Converting analog signals into digital
-

19.

NIC translates computer data into:

- A) Optical signals
 - B) Electrical signals
 - C) Radio waves
 - D) Infrared signals
-

20.

Modern motherboards usually have:

- A) No NICs
 - B) Built-in NICs
 - C) External NICs only
 - D) Wireless NICs only
-

✓ Answers with Explanations

1. **B** – Communication device = hardware transmitting/receiving data.
2. **B** – Switch connects computers and peripherals in a network.
3. **B** – Router determines optimal path and controls traffic.
4. **C** – Router = hardware + software.
5. **B** – Modem = Modulator + Demodulator.
6. **B** – Modulation = digital → analog.
7. **B** – Demodulation = analog → digital.
8. **D** – Modems use phone lines, cables, satellites.
9. **B** – Dial-up uses standard telephone lines.

- 10.**B** – Telephone lines carry analog signals.
- 11.**A** – DSL = Digital Subscriber Line.
- 12.**B** – DSL modems have built-in switches, higher speed.
- 13.**B** – Some DSL modems support wireless.
- 14.**A** – ISDN = Integrated Services Digital Network.
- 15.**C** – ISDN transmits data, voice, video simultaneously.
- 16.**B** – ISDN is faster but expensive.
- 17.**B** – NIC installed on motherboard.
- 18.**B** – NIC establishes physical connection.
- 19.**B** – NIC translates data into electrical signals.
- 20.**B** – Modern motherboards have built-in NICs.